

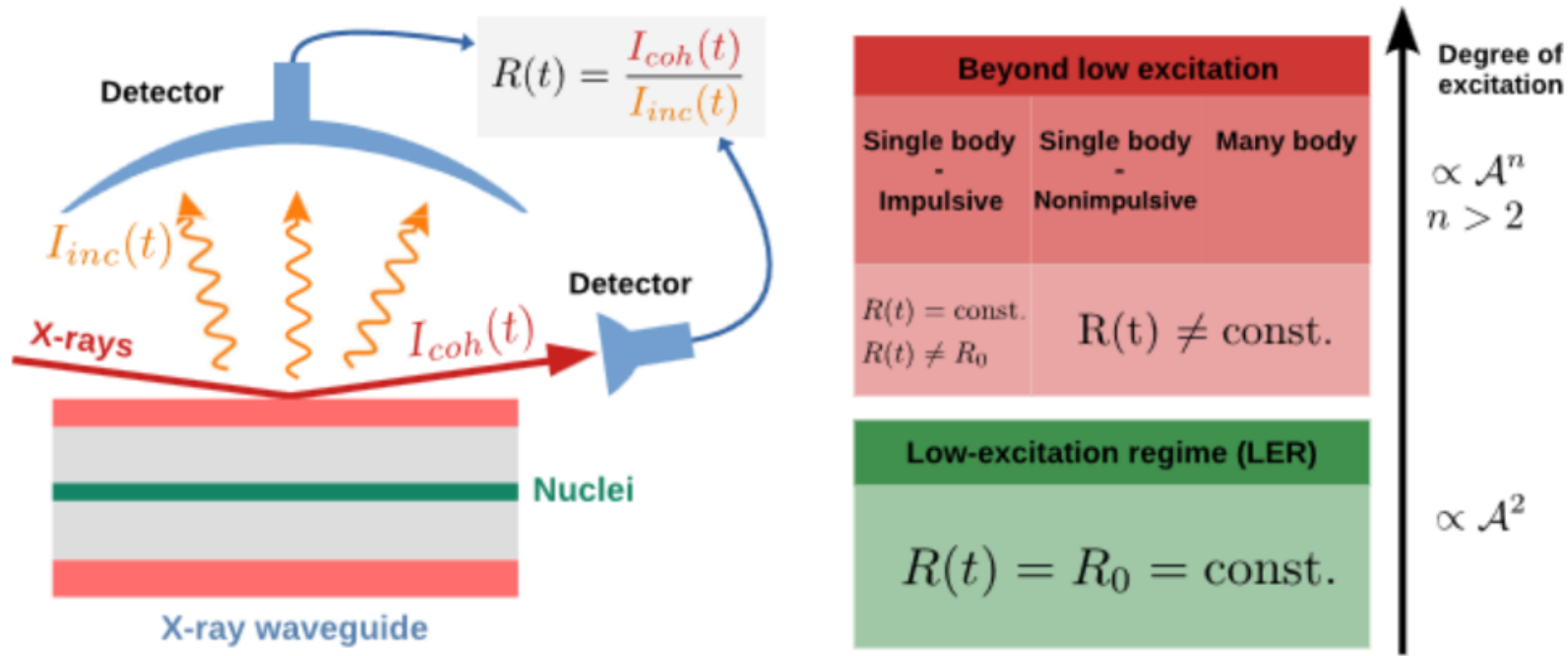


FIG. 1. Schematic setup and summary of the main results. The central goal of this work is to identify experimentally relevant signatures which enable one to verify the excitation of an ensemble of Mössbauer nuclei beyond the low-excitation regime (LER) explored up to now. For this, we consider an ensemble of two-level nuclei embedded in an x-ray waveguide, probed by near-resonant x-rays in grazing incidence. To detect the nonlinear excitation, the coherently and the incoherently (e.g., following internal conversion) scattered intensities are recorded. Our theoretical analysis shows that in the LER, the ratio $R(t) = R_0$ of the two intensities is constant, even though both intensities individually depend on time. We show this by analytically calculating the relevant dynamics of the excited-state populations and the x-ray-induced coherences of the general interacting N -body system in second order of the x-ray pulse area $\mathcal{A}$. Upon excitation beyond the LER, the ratio R changes its properties in a characteristic way. In case of near-instantaneous impulsive excitation of an effective single-particle system or a sufficiently weakly coupled many-body system, the ratio R remains constant, but changes its value from R_0 depending on the degree of excitation. For a strongly interacting impulsively driven many-body system, the ratio becomes time-dependent upon excitation beyond the LER. In case of nonimpulsive x-ray excitation with duration of the order of the nuclear lifetime, the ratio is also time dependent at higher excitation. Based on these results, a variety of different experimental signatures or data analysis approaches is developed, which allow one to verify excitations beyond the LER.